

Database Normalization

CIS 3730 — Lecture Handout

1. Introduction to Normalization

Database normalization is defined as the process of organizing columns and tables of a relational database to minimize data redundancy. The technique was introduced by Edgar Codd as part of his relational model. A poorly normalized schema stores the same fact in many places, which permits those copies to disagree with one another. An update anomaly occurs when changing a fact requires updating many rows and one is missed. An insertion anomaly is a situation where a fact cannot be recorded because some unrelated information is not yet known. A deletion anomaly is the loss of one fact as a side effect of deleting another.

2. Functional Dependency

A functional dependency is a constraint between two sets of attributes in a relation. Attribute B is functionally dependent on attribute A when each value of A is associated with exactly one value of B. A candidate key is a minimal set of attributes that functionally determines every other attribute in the relation. A partial dependency is a dependency on only part of a composite candidate key. A transitive dependency exists when a non-key attribute depends on another non-key attribute rather than on the key itself. Identifying functional dependencies correctly is the prerequisite for every normalization step that follows.

3. The Normal Forms

First normal form requires that every attribute contain only atomic values, so a column may not hold a list. Second normal form requires first normal form and the removal of every partial dependency on a composite key. Third normal form requires second normal form and the removal of every transitive dependency between non-key attributes. Boyce-Codd normal form is a stricter version of third normal form in which every determinant must be a candidate key. Each successive normal form eliminates a specific class of anomaly, and the forms are cumulative.

4. Decomposition and Trade-offs

Decomposition is the process of splitting one relation into several smaller relations. A decomposition is lossless when the original relation can be reconstructed exactly by joining the pieces back together. A dependency-preserving decomposition is one where every functional dependency can still be enforced without performing a join. Normalization optimizes for write correctness, not read speed. Analytical systems frequently denormalize deliberately, accepting redundancy in exchange for avoiding expensive joins at query time. The star schema is a denormalized design commonly used in data warehouses. Choosing a normal form is therefore an engineering trade-off rather than a rule to be followed blindly.